Reflected Intensity Correction Based on Global 3-D Plant Structure Acquired by Hyperspectral LiDAR

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***Abstract*—The precise quantification of 3-D plant biochemical parameters is essential for ecosystem assessment, yet acquiring accurate target reflectance remains a significant challenge for hyperspectral LiDAR (HSL) due to the strong incidence angle effect induced by complex leaf geometric morphology. Furthermore, existing correction methods are predominantly applied at a single-point scale rather than being strictly limited to it, struggling to accommodate the architectural complexity of whole plants. To address this bottleneck, this study proposes a novel reflectance correction framework for HSL based on the global 3-D structural information of the entire plant to suppress the effects of incidence angle on reflectance. Methodologically, an adaptive neighborhood point determination method based on point density is designed to accurately compute leaf curvature and normal vectors from point clouds, thereby avoiding over-smoothing or geometric distortion. Subsequently, a physically improved Cook–Torrance model for HSL (HSL-CT) is established to decouple diffuse and specular reflectance contributions across 28 spectral bands ranging from 445 to 810 nm. The evaluation results demonstrate that the proposed HSL-CT model consistently outperforms the baseline Lambert–Beckmann (LB) model in overall fitting accuracy and spectral fidelity. By integrating this model with the extracted global structural information, the framework effectively eliminates angular-induced specular over-brightness, significantly reduces spectral errors, and successfully restores native plant spectral signatures. Ultimately, this method can generate incidence-angle-corrected HSL reflectance data with high stability, laying a solid foundation for accurate inversion of biochemical parameters at the individual plant scale.**

***Index Terms*—3-D plant reflection, hyperspectral LiDAR (HSL), incident angle effect, radiation modeling.**

I. Introduction

THE precise quantification of 3-D plant biochemical parameters via remote sensing, which is essential for monitoring growth and health, underpins critical efforts in ecosystem assessment, crop yield estimation, and carbon cycle modeling [1], [2], [3]. In conventional plant remote sensing, hyperspectral imaging enables the acquisition of detailed spectral data yet lacks accurate 3-D spatial information [4], [5]. Conversely, while single-wavelength LiDAR can retrieve high-precision spatial attributes, it typically operates at a single wavelength, thereby limiting its spectral discrimination capability [6], [7]. Although multimodal techniques attempt to integrate the merits of both sensors, they frequently confront substantial registration complexities and struggle to accurately capture understory spectra when concurrently utilizing disparate datasets [8], [9]. Driven by recent advancements in LiDAR hardware, hyperspectral LiDAR (HSL) has emerged as a revolutionary active optical technology that seamlessly couples the spatial resolution of conventional LiDAR with the superior spectral fidelity of hyperspectral imaging [10], [11], [12]. By simultaneously acquiring and generating 3-D point clouds enriched with multi-wavelength intensity information, HSL offers a transformative paradigm for individual tree segmentation, species identification, and biochemical parameter inversion [13], [14], [15].

The precise acquisition of target reflectance remains a primary bottleneck in quantitative HSL applications. While the distance effect in HSL is independent of target properties and relatively straightforward to mitigate, the incidence angle effect is inherently coupled with target surface attributes and presents a far greater correction challenge [16]. This difficulty arises because the intricate geometric morphology of plant leaves induces highly anisotropic and volatile scattering behaviors in the backscattered signals [17]. These bidirectional reflection components are not only regulated by leaf biochemical properties but are also highly sensitive to variations in observation geometries and scanning configurations [18]. When directly incorporated into quantitative inversion models, such uncorrected fluctuations disrupt the intrinsic physical correspondence between spectral signatures and plant parameters,

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a critical issue prevalent across both passive and active remote sensing domains [19], [20], [21], but particularly pronounced in HSL [22]. To alleviate this problem, Xu et al. [15] used an empirical approach based on the cosine law of incidence angle. However, this strategy relies on an idealized Lambertian assumption, thereby yielding unsatisfactory performance for target leaves that exhibit pronounced specular reflections and substantial angle-induced brightness variations. In contrast, earlier research [23], [24] utilized multiangle leaf datasets based on the Lambert–Beckman model, demonstrating its effectiveness in correcting angular effects. Zhang et al. [25] developed a Lambert–Beckmann (LB) exponential model that accounts for laser spot distribution, achieving excellent consistency between the corrected return intensity and the intensity measured in the normal direction.

The prominent advantage of HSL lies in its unique capability to simultaneously capture high-resolution spatial structures and fine spectral signatures. However, state-of-the-art HSL hardware constraints restrict the spatial resolution of individual-plant point clouds to a typical point spacing of approximately 5 mm [26], [27]. Although this sampling density adequately supports structural classifications, canopy segmentation, or statistical machine learning inversions, it introduces nonnegligible geometric uncertainty when resolving microscale incidence angle effects across complex 3-D canopies. This discrete sampling characteristic directly hinders the reconstruction of accurate leaf curvature and normal vectors, thereby leaving virtually all existing incidence angle correction methodologies predominantly focused on the idealized single-point or single-leaf scale [23], [28]. Whether such point-based methods can be effectively scaled to accommodate the intricate architectural complexity of individual plants remains a subject of ongoing debate and requires rigorous validation. For potted plants, leaf distribution and 3-D morphology display a high degree of heterogeneity, far exceeding the complexity observed at the single-point scale. Effectively mitigating incidence angle effects at the whole-plant scale to achieve accurate reflectance retrieval, by fully incorporating this inherent structural complexity, remains a formidable challenge in quantitative HSL remote sensing.

To fill this research gap, this study adopts high-precision point cloud data captured by a ground-based HSL system to analyze the internal mechanism linking global 3-D plant structures to HSL angular bias. The main contributions of this article are listed below.

1) An integrated correction workflow is established to mitigate incidence angle bias during HSL reflectance retrieval. Distinct from conventional single-point correction strategies, this framework incorporates holistic 3-D structural information of individual plants for radiometric calibration.

2) An adaptive neighborhood point selection method driven by local point density is proposed. The approach mitigates oversmoothing and geometric distortion artifacts induced by spatially uneven point sampling during leaf surface normal and incidence angle calculation.

3) A physically revised Cook–Torrance model tailored for HSL (HSL-CT) is constructed. The model is validated quantitatively using multiangle measurement datasets covering multiple plant leaf species with varying incidence angles.

4) High-fidelity HSL point cloud data of individual plants are presented, achieving the highest level of visual quality and spectral consistency reported to date.



Fig. 1. Overview of potted plant and leaf datasets. (a) Experimental scenario for potted plant. (b) Close-up view of the potted plant morphology. (c) Representative leaf samples across different plant species. (d) Experimental setup for multiangle leaf data collection.

II. Materials and Methods

A. Experiment Installation and Plant Datasets

This study utilizes 3-D point clouds of whole plants acquired via terrestrial HSL and constructs two plant datasets to validate the comprehensive performance of the proposed correction framework. Both datasets cover a spectral range of 445–810 nm, comprising 28 spectral bands with a spectral bandwidth of 13 nm per band. Both acquisition experiments were conducted at a fixed scanning range of 5.8 m, which yields a laser footprint diameter of 6.69 mm, and employed the LiDAR equation to eliminate distance-induced intensity variations caused by target depth. The first dataset consists of point cloud data from a potted *Anthurium* plant acquired using a high-precision scanning unit, containing approximately 170 000 points with an average spatial point spacing of 1.40 mm perpendicular to the laser line of sight. For the potted plant data, continuous whiteboard echo intensities were collected within a distance range of 1–10 m. Combined with the physical principle that lidar echo intensity is inversely proportional to the square of the scanning distance, we implemented a dedicated distance effect correction to eliminate echo intensity discrepancies of leaves located at different depth positions of the potted plant. This dataset serves to evaluate and validate the overall methodological framework, as illustrated in Fig. 1(a) and (b) for the potted plant setup. The second dataset comprises multiangle single-point measurements of detached leaves, which are specifically used for the evaluation of reflectance properties modeling. Observations

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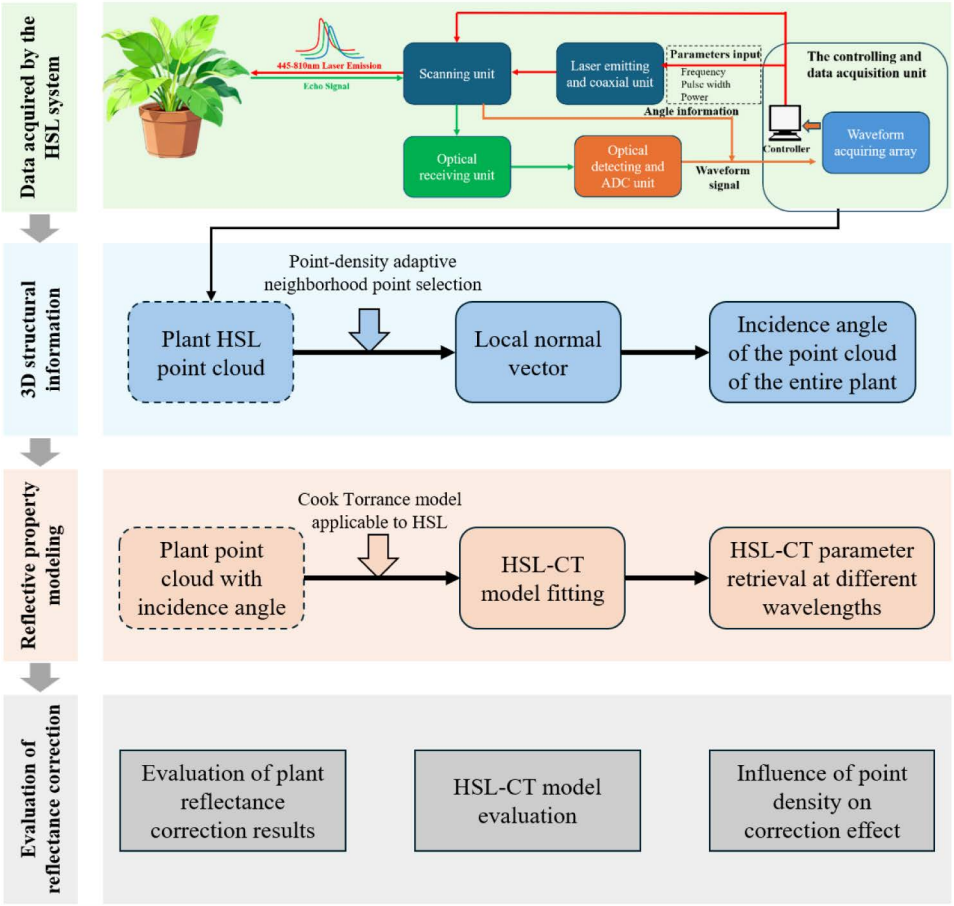


Fig. 2. Flowchart of the 3-D HSL framework for plant reflectance correction.

were conducted on leaves from 14 distinct plant species across incidence angles ranging continuously from 0° to 76° at 4° increments, while synchronously recording spectral intensities across the 28 bands, as depicted in Fig. 1(c) and (d). A standard white reference panel with 99% reflectance was scanned synchronously throughout the experiment for relative radiometric calibration, thereby converting raw measurements into true leaf reflectance.

B. Framework of Reflectance Correction Based on Global 3-D Plant Structural Information

As illustrated in Fig. 2, this study utilizes a previously developed terrestrial HSL system to acquire hyperspectral point cloud data for plants [29]. The system consists of a laser emitting and coaxial unit, a scanning unit, an optical receiving unit, an optical detection and analog-to-digital converter unit, and a controlling and data acquisition unit. This controlling and data acquisition unit includes a controller and a waveform acquiring array. As depicted, the laser emitting and coaxial unit undertakes laser emission and allows for parameter input like frequency, pulsewidth, and power. The nominal laser beam divergence angle provided by the manufacturer is less than 2 mrad. The scanning unit accomplishes the laser scanning action. The optical receiving unit receives the echo signal reflected by the target, such as plants. The optical detection and ADC unit conducts detection on the laser signal and performs analog-to-digital conversion to generate a waveform signal. The controlling and data acquisition unit is utilized for parameter input, waveform signal acquisition, and overall system control, with all these units collaborating via signal transmission and control instructions to achieve laser detection and data acquisition of the target [30].

Based on data acquired via the terrestrial HSL system, this study develops a 3-D correction framework for whole potted plants to eliminate incidence angle effects. The detailed technical workflow is illustrated in Fig. 2, comprising three main stages: 3-D structural parameter extraction, reflectance property modeling, and incidence angle effect correction. The first stage focuses on extracting 3-D structural parameters: taking the hyperspectral point cloud of the potted plant as input, the framework adaptively selects an optimal neighborhood to calculate robust normal vectors, thereby determining the laser incidence angle at each leaf surface point. The second stage addresses research on reflectance property modeling: based on the HSL backscattering mechanism, we develop an improved HSL-CT model tailored for HSL applications. For potted plants, their reflectance property modeling is completed using a single leaf of the same plant. The comprehensive evaluation and method comparison of the HSL-CT model are performed on 14 leaf samples across different cultivars and multiangle measurement settings. This model effectively

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isolates the specular component from the diffuse component at any arbitrary incidence angle. The final stage executes the incidence angle effect correction: by integrating the calculated leaf surface incidence angles with the complete set of Cook–Torrance parameters, the system computes the specular reflection ratio across varying incidence angles, ultimately yielding refined radiometric plant data.

C. 3-D Structure Information Calculation

For both terrestrial single-wavelength and HSL systems, point clouds captured from nominally flat surfaces, such as leaves or walls, inevitably display an apparent thickness effect, characterized by localized spatial dispersion along the depth axis rather than an ideal 2-D planar distribution [31], [32]. Furthermore, this thickness is closely related to the hardware system, physical scattering, and scanning geometry, rather than being a single, fixed value. Relying solely on a fixed-neighborhood principal component analysis (PCA) method to calculate surface normals leads to issues such as over-smoothing or excessive roughness. To address this, we propose an adaptive method for neighborhood selection and normal vector calculation that explicitly accounts for point cloud density (as shown in Method 1). The method first transforms the 3-D point cloud into a spherical projection relative to the scanning viewpoint, discretizing the space into a 2-D physical grid to derive the global mean density ρ_{avg} and eliminate range-induced nonuniformity. Building upon this baseline, the algorithm enters a fixed-point iterative optimization phase: in each iteration, the current neighborhood size K_{curr} is used to estimate local surface normals via PCA, while points within each grid cell are projected onto the normalized mean normal vector to extract the topological thickness T_{avg}. A target neighborhood size K_{next} for the subsequent iteration is then dynamically derived by combining T_{avg}, a smoothing multiplier α, and the density prior. Upon reaching the convergence tolerance ΔK_{tol}, the system settles on the optimal neighborhood size K_{opt} and robust normal vectors n_{opt}, which are subsequently used to compute accurate incidence angles θ for downstream radiometric correction models.

Three key parameters of the algorithm (α multiplier, $gridSize$, and K_{init}) are optimized via a grid search strategy. Other parameters are empirically fixed to ensure numerical stability and algorithmic safety: the convergence tolerance ΔK_{tol} is set to 20 to guarantee stable fixed-point convergence while mitigating minor numerical oscillations; concurrently, the maximum neighborhood bound K_{max} is thresholded at 1000 as a safeguard against unconstrained neighborhood expansion. For the parameter optimization process, we employ local normal consistency (LNC) and the fitting root-mean-square error (Fitting RMSE) to evaluate the results obtained from different parameter combinations. LNC is primarily used to quantify the algorithm's ability to suppress "thickness noise." Within an infinitesimal neighborhood, the surface of a real plant leaf should be continuous and smooth. If the selected neighborhood size is overly small, noise will severely distort estimated normals and produce random normal orientations. This further causes spatially inconsistent normal directions

Method 1 Adaptive Neighborhood Selection

Input: P(point cloud[X, Y, Z]), $viewpoint$, $gridSize$, α(multiplier), K_{init}, K_{max}, ΔK_{tol}

Output: $P_{refined}$(point cloud with robust normal n_{opt} and incidence angles θ), K_{opt}

Begin

Load point cloud coordinates [X, Y, Z] from P

//Step 1: Calculate viewpoint projection grid density

Calculate point distances $r \leftarrow \sqrt{X^2 + Y^2 + Z^2}$

$R_{mean} \leftarrow \text{mean}(r)$

Convert (X, Y, Z) to spherical projection coordinates (ϕ, θ_{sph}) relative to *viewpoint* Map to 2D physical projection space:

$proj_X \leftarrow \phi \times \cos(\theta_{sph}) \times R_{mean}$

$proj_Y \leftarrow \theta_{sph} \times R_{mean}$

Assign points to projection grids:

$grid_X \leftarrow \text{floor}(proj_X / gridSize)$

$grid_Y \leftarrow \text{floor}(proj_Y / gridSize)$

Count points in each unique $(grid_X, grid_Y)$ cell $\rho_{avg} \leftarrow$ average point count across all nonempty cells

Group points in P by grid cell indices $\rightarrow P_{grouped}$

//Step 2: Fixed-point iteration for adaptive K

$K_{curr} \leftarrow K_{init}$

$converged \leftarrow False$

While not *converged*:

Perform neighborhood detection via KD-Tree with K_{curr}

$n_{curr} \leftarrow$estimate normal vectors using PCA for all points

Orient all n_{curr} toward *viewPoint*

Group normal n_{curr} by grid cell indices $\rightarrow N_{grouped}$

Initialize $thicknessList \leftarrow \emptyset$

For each cell (p_cell, n_cell) in $(P_{grouped}, N_{grouped})$:

If size$(p_cell) \geq 2$:

$n_{mean} \leftarrow mean(n_cell)/|\text{mean}(n_cell)|$

$projections \leftarrow \text{dot_product}(p_cell, n_{mean})$

$cellThickness \leftarrow \max(projections)$

-min($projections$)

Append $cellThickness$ to $thicknessList$

$T_{avg} \leftarrow mean(thicknessList)$

Calculate target neighborhood size:

$targetArea \leftarrow (T_{avg} \times \alpha)^2$

$K_{next} \leftarrow integer(targetArea \times \rho_{avg})$

$K_{next} \leftarrow min(K_{next}, K_{max})$

If $|K_{next} - K_{curr}| \leq \Delta K_{tol}$:

$converged \leftarrow True$

$K_{opt} \leftarrow K_{next}$

$n_{opt} \leftarrow n_{curr}$

Else:

$K_{curr} \leftarrow K_{next}$

//Step 3: Compute final incidence angles

For each point P_i and normal $n_{opt,i}$:

$\cos(\theta_i) \leftarrow |P_i \cdot n_{opt,i}| / \|P_i\|$

$\theta_i \leftarrow arccos(clip(\cos(\theta_i), -1, 1))$

Save $[P, n_{opt}, \theta]$ to $P_{refined}$

End

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and large angular deviations between neighboring normal vectors. When parameter tuning effectively reduces range noise, adjacent normals become spatially coherent, accompanied by a notable drop in LNC. LNC is calculated as follows:

$$LNC = \frac{1}{Nk}\sum_{i=1}^{N}\sum_{j=1}^{k}\arccos\left(|n_i \cdot n_{i,j}|\right). \tag{1}$$

Here, N denotes the total number of points within the point cloud, and k represents the number of neighboring points used to evaluate local consistency, which is empirically set to 30. The term n_i designates the optimized normal vector at point P_i, whereas $n_{i,j}$ represents the surface normal of the jth neighbor within the defined local neighborhood. By computing the mean absolute angular deviation among these normal vectors, LNC provides a robust quantitative metric to evaluate local surface smoothness following the suppression of range noise.

Fitting root mean square error (RMSE) is primarily used to determine whether the algorithm causes "over-smoothing," thereby assessing the fidelity of geometric curvature preservation. If the neighborhood range introduced during iteration is excessively large and covers the natural curvature of the leaf surface, the fitted local plane will fail to conform closely to the actual leaf surface, causing a sharp increase in the perpendicular distances from points within the neighborhood to that plane. RMSE is calculated as follows:

$$RMSE = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{1}{K}\sum_{j=1}^{K} d_{i,j}^2}. \tag{2}$$

Here, K represents the optimal neighborhood size obtained upon the convergence of the adaptive algorithm. The variable $d_{i,j} = (P_{i,j} - \overline{P}_i) \cdot n_i$ denotes the orthogonal distance from the jth neighboring point $P_{i,j}$ to the locally fit plane, where $\overline{P}_i$ signifies the geometric centroid of the local neighborhood. This metric quantifies the spatial deviation between the estimated plane and the actual point cloud morphology, wherein the RMSE increases sharply if an excessively large value of K induces oversmoothing.

To eliminate dimensional discrepancies between LNC and RMSE and to objectively quantify the trade-off between local smoothing and feature fidelity, a comprehensive geometric cost function denoted as CGC is introduced for parameter optimization as follows:

$$CGC = \frac{\widehat{LNC} + \widehat{RMSE}}{2}. \tag{3}$$

First, Min-Max normalization is applied to map both LNC and RMSE onto a normalized scale from 0 to 1, yielding the normalized metrics $\widehat{LNC}$ and $\widehat{RMSE}$. Subsequently, the CGC metric quantifying overall geometric quality is constructed by averaging these normalized values. A lower CGC value indicates that the current parameter set effectively suppresses range noise while maximally preserving the true leaf surface curvature. Consequently, the optimal parameter solution for the adaptive algorithm corresponds to the global minimum of the CGC curve.

D. Reflective Property Modeling

1) LB Model: The LB model, adopted by Bai et al. [23], has been widely applied and recognized as *a* primary baseline method for eliminating the incidence angle effect in HSL data processing. Based on geometric optics, this model decomposes the overall reflectance into two additive components: an isotropic diffuse scattering term obeying Lambert's cosine law, and *a* directional specular reflection term governed by the Beckmann distribution. For HSL, the model expresses the total reflectance f_r at a given incidence angle θ as follows:

$$f_r = I_0\left[k_d\cos\theta + (1-k_d)\frac{\exp\left(-\frac{\tan^2\theta}{\alpha^2}\right)}{\cos^5\theta}\right] \tag{4}$$

where θ represents the incidence angle between the target surface normal vector and the incident direction vector. In this equation, the first term $k_d\cos\theta$ describes the diffuse reflection component, whereas the second term represents the specular reflection component. The physical meanings of the key parameters are defined as follows: I_0 denotes the normal backscattered intensity scaling factor, k_d ($0 \leqslant k_d \leqslant 1$) signifies the proportion of diffuse scattering relative to total reflection, and α is the target surface roughness parameter. Among these variables, three key parameters, specifically I_0, k_d, and α, must be calibrated for each spectral band via nonlinear fitting using multiangle experimental datasets acquired from leaf samples. In this study, this method is adopted as a benchmark to evaluate the performance and demonstrate the superiority of the proposed HSL-CT model.

2) Proposed HSL-CT Model: Similar to the LB model, the Cook–Torrance model reconstructs the true reflection characteristics of the target by decomposing the total reflection into diffuse and specular components while integrating microfacet theory and physical laws [33]. The mathematical expression of the echo intensity f_r at a point on the target surface is shown as follows:

$$f_r = \frac{(1-F_{avg})\rho_d}{\pi} + \frac{F(\omega_i,h)\,D(h,n,\alpha)\,G(\omega_i,\omega_o,n,\alpha)}{4(n\cdot\omega_o)(n\cdot\omega_i)}. \tag{5}$$

Herein, F_{avg} denotes the average Fresnel reflectance, with $(1 - F_{avg})$ serving to prevent the total reflected energy from exceeding the incident light energy, and ρ_d is the true reflectance of the target. The key parameters are defined as follows: ω_i is the incident light direction vector, ω_o is the viewing direction vector, n is the target surface normal vector, h is the halfway vector (i.e., the angle bisector direction between the incident and viewing directions), and α is the target surface roughness parameter. Specifically, $D(h,n,\alpha)$ is the normal distribution function, which primarily counts the proportion of microfacets with normal directions aligned with the halfway vector h (determining the number of microfacets participating in reflection) and is calculated using the GGX distribution as follows:

$$D(h,n,\alpha) = \frac{\alpha^2}{\pi\left[(n\cdot h)^2(\alpha^2-1)+1\right]^2}. \tag{6}$$

$G(n,\omega_i,\omega_o,\alpha)$ is the geometric shadowing term, depicting the proportion of microfacets unobstructed by neighboring

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microfacets that allow incident light to reach and reflected light to exit (determining the proportion of effective reflective microfacets), with its calculation provided as follows:

$$G(n,\omega_i,\omega_o,\alpha)=\frac{2(n\cdot\omega_i)}{(n\cdot\omega_i)+\sqrt{\alpha^2+(1-\alpha^2)(n\cdot\omega_i)^2}}\cdot\frac{2(n\cdot\omega_o)}{(n\cdot\omega_o)+\sqrt{\alpha^2+(1-\alpha^2)(n\cdot\omega_o)^2}}. \quad (7)$$

$F(\omega_i,h)$ is the Fresnel term, characterizing the ratio of reflected light to incident light at various angles (determining the reflected energy proportion of effective microfacets) and can be computed via Schlick's approximation as follows, where F_0 is the normal Fresnel reflectance:

$$F(\omega_i,h)=F_0+(1-F_0)\cdot(1-h\cdot\omega_i)^5. \quad (8)$$

The echo signal of HSL is a backscatter signal, where the laser incident direction is consistent with the viewing direction (i.e., $\omega_i=\omega_o$). In the Cook–Torrance model, F_0, F_{avg}, α and ρ_d are theoretically dependent on the wavelength k. Thus, the Cook–Torrance model suitable for N bands HSL (HSL- CT model) can be represented by (8) and (9), with θ denoting the laser incident angle

$$\begin{aligned}&f_r(k)=\frac{\left(1-F_{avg}(k)\right)\rho_d(k)}{\pi}+\frac{F_{back}(k)D_{back}(\theta,\alpha)G_{back}(\theta,\alpha)}{4\cos^2\theta}\\&k=1,2,3\ldots,N F_{back}(k)=F_0(k)\end{aligned} \quad (9)$$

$$\begin{cases}D_{back}(\theta,\alpha)=\dfrac{\alpha^2}{\pi\cdot\left[\cos^2\theta\cdot(\alpha^2-1)+1\right]^2}\\ G_{back}(\theta,\alpha)=\left(\dfrac{2cos\theta}{cos\theta+\sqrt{\alpha^2+(1-\alpha^2)cos^2\theta}}\right)^2\end{cases}. \quad (10)$$

Here, the first term of $f_r(k)$ represents the isotropic diffuse reflection component, whereas the second term of $f_r(k)$ denotes the directional specular reflection component. By calculating the ratio of the first term to the total backscattered reflectance $f_r(k)$, the framework effectively suppresses specular interference and eliminates geometry-induced radiometric distortions.

Based on the dataset of multiangle single-point leaf observations spanning incidence angles from 0° to 76° at 4° intervals across 28 spectral bands, nonlinear least-squares fitting was employed to determine the optimal set of model parameters, specifically $\rho_d(k)$, $F_{avg}(k)$, $F_0(k)$, and α. The fitting loss function is defined as the mean squared error between the observed reflectance and the model-predicted reflectance, expressed as follows:

$$\mathcal{L}(k)=\sum_{\theta}\left[\hat{f}_r(k,\theta)-f_r\left(k,\theta,\rho_d,F_{\text{avg}},F_0,\alpha\right)\right]^2. \quad (11)$$

Here, $\mathcal{L}(k)$ represents the sum of squared residuals corresponding to the kth spectral band, while $\hat{f}_r$ denotes the measured normalized echo intensity value. Upon convergence of the optimization process, a comprehensive HSL-CT model parameter library covering all 28 HSL spectral bands and demonstrating broad applicability across diverse leaf species is established.

For each point within the global point cloud of the potted plant, the method described in Section II-C is applied to determine the incidence angle. By combining this with the complete set of HSL-CT parameters fit for the specific spectral band, the proportion of specular reflection w_{spec} across 28 spectral bands can be calculated as follows:

$$w_{\text{spec}}(k,\theta)=\frac{\frac{F_{back}(k)D_{back}(\theta,\alpha)G_{back}(\theta,\alpha)}{4\cos^2\theta}}{f_r(k)}. \quad (12)$$

Therefore, the reflectance ρ_{corr} following the elimination of specular reflection is calculated as follows:

$$\rho_{\text{corr}}(k,\theta)=\rho_{\text{raw}}(k,\theta)\cdot\left[1-w_{\text{spec}}(k,\theta)\right]. \quad (13)$$

Here, $\rho_{\text{corr}}(k,\theta)$ represents the corrected reflectance following the suppression of specular interference at the given incidence angle, while $\rho_{\text{raw}}(k,\theta)$ denotes the raw reflectance measured by the terrestrial HSL system.

E. Evaluation Metrics

To quantitatively evaluate the performance of the proposed reflectance correction method and the fitting precision of the model, several primary statistical metrics are utilized. spectral angle mapper (SAM) is employed to assess the spectral similarity between the highlight regions and the reference regions before and after correction. A smaller SAM value indicates higher spectral similarity and superior correction of the incidence angle effect. The standard deviation (Std) and its reduction rate serve to evaluate the consistency of the corrected reflectance within a leaf. A lower Std and a higher reduction rate signify superior correction performance. The coefficient of determination R^2 characterizes the correlation and degree of agreement between datasets. The value closer to 1 represents higher data reliability. The RMSE quantifies the absolute deviation between the corrected reflectance and the reference reflectance. This metric provides a direct measure of the accuracy of reflectance restoration across different spectral bands

$$SAM=\cos^{-1}\left(\frac{\sum_{i=1}^{n}\rho_{cor,i}\rho_{ref,i}}{\sqrt{\sum_{i=1}^{n}\rho_{cor,i}^2}\sqrt{\sum_{i=1}^{n}\rho_{ref,i}^2}}\right) \quad (14)$$

$$Std=\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\rho_{cor,i}-\bar{\rho}_{cor}\right)^2} \quad (15)$$

$$R^2=1-\frac{\sum_{i=1}^{n}\left(\rho_{ref,i}-\rho_{cor,i}\right)^2}{\sum_{i=1}^{n}\left(\rho_{ref,i}-\bar{\rho}_{ref}\right)^2} \quad (16)$$

$$RMSE=\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\rho_{cor,i}-\rho_{ref,i}\right)^2}. \quad (17)$$

In these equations, n represents the total number of spectral bands or angular measurements, while $\rho_{cor,i}$ and $\rho_{ref,i}$ denote the corrected reflectance and the reference ground truth reflectance at the ith band, respectively, with $\bar{\rho}_{ref}$ being the mean value of the reference reflectance. For the spatial consistency evaluation, N denotes the total number of evaluated

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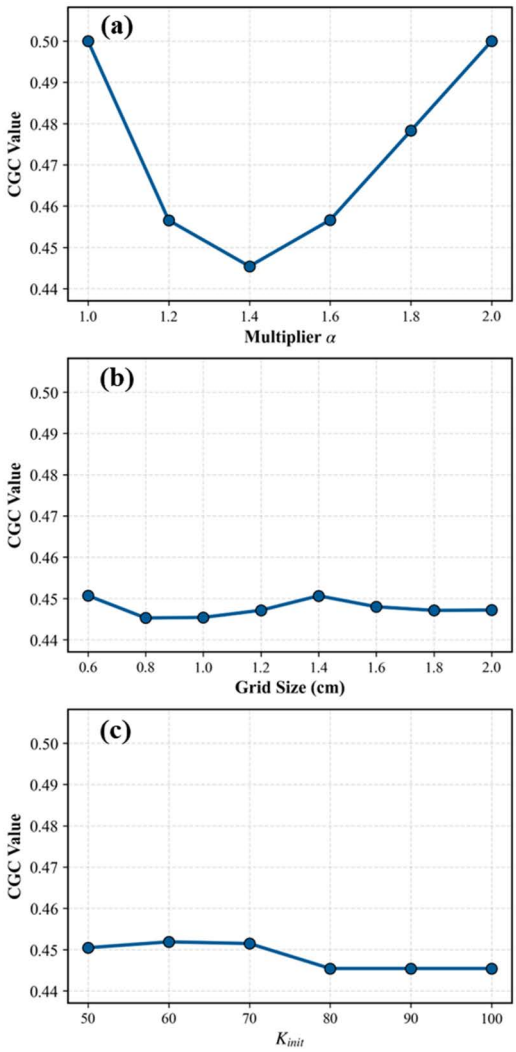


Fig. 3. Parameter optimization and robustness analysis based on the CGC. (a) Geometric trade-off analysis of the scaling multiplier. (b) Scale robustness analysis of the initial grid size. (c) Convergence stability analysis of the neighborhood size.

points within a specific leaf region, where $\rho_{cor,i}$ represents the corrected reflectance of the ith point, and $\overline{\rho}_{cor}$ is the mean corrected reflectance of all points in that region.

III. Results

A. Global 3-D Plant Structure Result

Grid parameter optimization was performed to determine the optimal values for the Multiplier, grid size, and K_{init}, with the optimization results illustrated in Fig. 3. The optimal parameter combination was identified as a multiplier of 1.4, a grid size of 0.8, and a K_{init} of 80. Based on this optimal configuration, the number of neighboring points selected for calculating normal vectors and incidence angles was determined to be 744.

In the experimental setup, the potted plant was placed diagonally below the HSL system. We calculated the spatial coordinates of the points using the laser emission point as the origin. Consequently, the angle between the leaf normal vector and the vector pointing back to the origin remains acute. Fig. 4 illustrates the overall normal vector distribution for the potted plant. Specifically, Fig. 4(a) displays the true-color (RGB) 3-D point cloud, illustrating the actual plant morphology, while Fig. 4(b)–(d) depict the 3-D spatial distributions of the normal vector components along the X-, Y-, and Z-axes, respectively. The color scale for all components ranges from −1 to 1, facilitating an intuitive visual assessment of normal vector magnitudes. Notably, most Z-axis components in Fig. 4(d) are positive (indicated by red and yellow areas), which aligns with the natural upward orientation of the leaves.

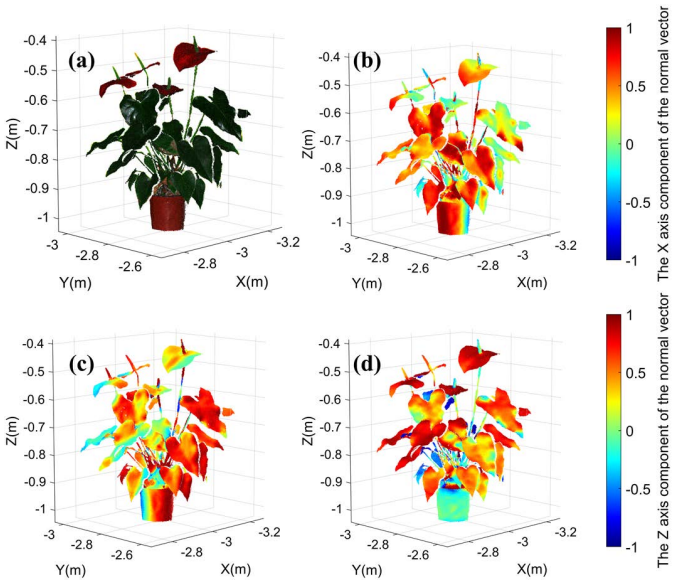


Fig. 4. Normal vector distribution of the plant canopy. (a) True-color 3-D point cloud. (b) X-axis component. (c) Y-axis component. (d) Z-axis component.

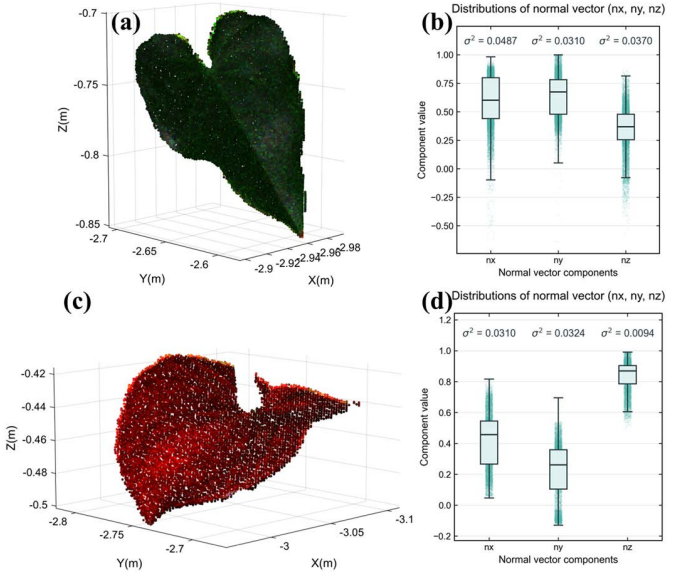


Fig. 5. Normal vector distributions and variances for leaves in varying orientations. (a) 3-D point cloud of a leaf with a forward-facing upper surface. (b) Boxplots and variances of normal vector components for leaf (a) and (c) 3-D point cloud of a leaf with an upward-facing upper surface. (d) Boxplots and variances for leaf (c).

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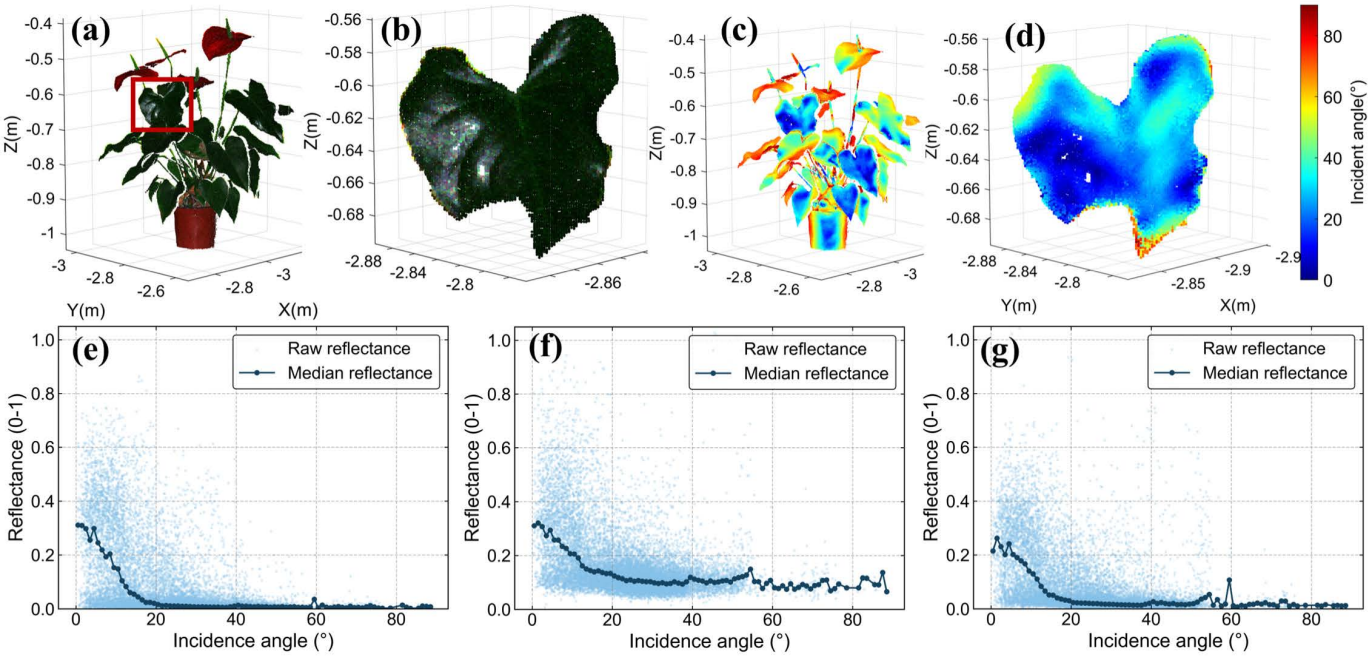


Fig. 6. Incidence angle distribution and its correlation with reflectance for the plant point cloud. (a) 3-D true-color (RGB) point cloud of the entire plant. (b) Enlarged true-color view of the target leaf. (c) 3-D visualization of incidence angles. (d) Enlarged incidence angle visualization for the target leaf. (e)–(g) Raw and median reflectance versus incidence angle for the target leaf at 450.2, 546.3, and 654.3 nm, respectively.

Furthermore, the spatial distribution of all normal vector components is remarkably smooth, lacking abrupt transitions. This smoothness validates the accuracy and robustness of our calculation method, confirming its consistency with actual leaf orientations.

Since obtaining ground-truth normal vectors for canopy leaves is difficult, we statistically analyzed their distributions. Leaves with low curvature approximate smooth, continuous surfaces, yielding highly consistent normal vectors with narrow component variations. Therefore, we used variance to quantify this dispersion. Fig. 5 presents the normal vector statistics for two leaves in different orientations. Fig. 5(a) and (c) display the 3-D point clouds, illustrating the spatial morphology of the leaves. Fig. 5(b) and (d) show boxplots of the normal vector components, including labeled variances for quantitative analysis. For leaf (a), the variances (and estimated medians) of the X, Y, and Z components are 0.0487 (0.60), 0.0310 (0.68), and 0.0370 (0.37), respectively. For leaf (c), the corresponding values are 0.0310 (0.46), 0.0324 (0.26), and 0.0094 (0.87).

After determining the normal vectors for all plant canopy points, we calculate the incidence angles. This angle is formed between the normal vector and the laser incidence vector. We derive the laser incidence vector directly from the spatial coordinates of the points. Fig. 6 presents the incidence angle distributions and their correlations with return intensity. Each subplot highlights distinct features and variation patterns. Fig. 6(a) displays the 3-D true-color (RGB) canopy point cloud. A red bounding box highlights a specific target leaf. Fig. 6(b) provides an enlarged view of this selected leaf. Fig. 6(c) visualizes the incidence angles for all canopy points, with a color scale spanning 0°–90°. These gradients allow for an intuitive assessment of angle variations across different leaf positions. The spatial distribution of incidence angles aligns with actual leaf orientations; leaves facing the laser origin exhibit smaller incidence angles. Fig. 6(d) presents the corresponding enlarged incidence angle visualization for the target leaf. Fig. 6(e)–(g) plot reflectance against incidence angle for the target leaf at 450.2, 546.3, and 654.3 nm, respectively. In each subplot, light blue scatter points represent the raw reflectance data, while the dark blue line indicates the median reflectance. All three plots reveal a clear trend: the median reflectance decreases as the incidence angle increases. This attenuation is notably faster at smaller angles (<30°). This phenomenon aligns perfectly with established backscattering theories regarding reflectance variations. Consequently, this consistency indirectly validates the accuracy of our proposed incidence angle calculation method.

B. HSL-CT Model Fitting and Reflectance Correction Results

Having accurately determined the specific incidence angle for each point on the potted plant point cloud, the theoretical nonspecular reflectance at the corresponding angle must be ascertained. To this end, an individual leaf sampled from the same potted plant was utilized to calibrate the HSL-CT model parameters across all 28 wavelengths.

Fig. 7 presents the HSL-CT model parameter estimation results across multiple spectral bands. Subplots Fig. 7(a)–(j) show the fitting results for wavelengths ranging from 450.2 to 809.3 nm. In each subplot, blue scatter points represent the raw experimental data collected at various incidence angles. These data were globally normalized across all spectral channels. The red solid curve denotes the fit model. Each panel annotates the R^2 and the estimated parameters (ρ_d, F_{avg}, F_0, and α). These metrics allow for a quantitative assessment of model

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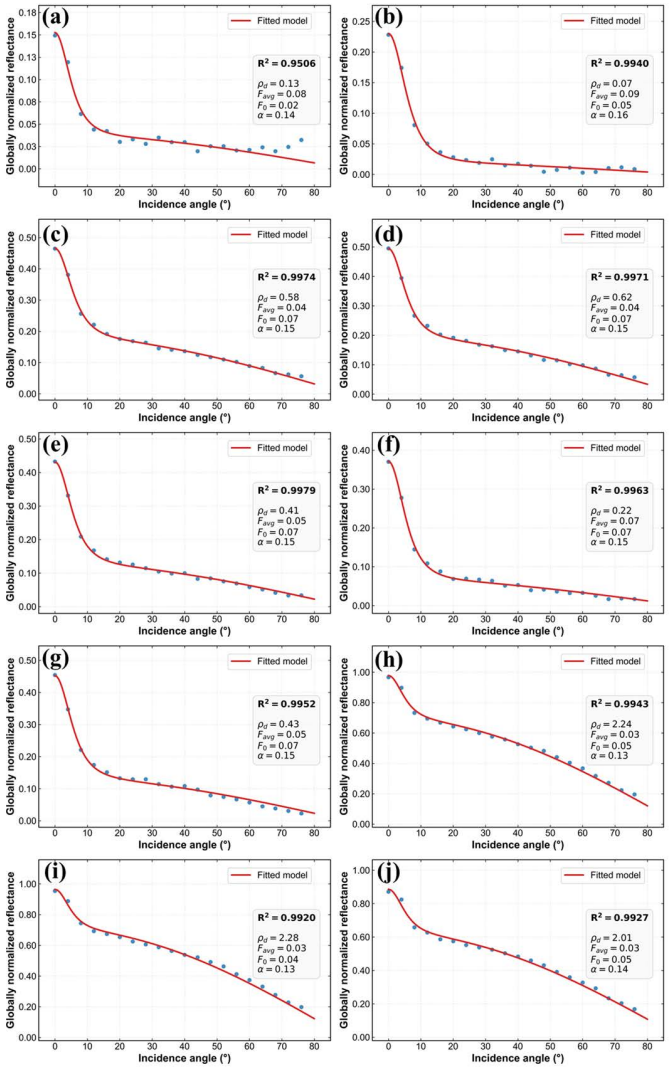


Fig. 7. HSL-CT model fitting results of globally normalized reflectance for potted plant leaves across various wavelengths. (a)–(j) Fitting results at 450.2, 491.5, 532.6, 573.5, 614.2, 654.3, 694.0, 733.2, 771.6, and 809.3 nm, respectively. In each subplot, blue scatter points denote the experimental data, while the red solid line represents the fit HSL-CT model. The R^2 and the specific fit parameters (ρ_d, F_{avg}, F_0, and α) are annotated within each panel.

performance and parameter features across different bands. Notably, at small incidence angles (typically <20°), the globally normalized reflectance increases rapidly as the incidence angle decreases. This reflects the typical characteristic of significantly enhanced HSL reflection intensity at small incidence angles on the leaf surface. Conversely, at larger incidence angles (>20°), the reflectance exhibits Lambertian scattering characteristics. The model maintains a high quality of fit across the entire angular range. This alignment with HSL incidence angle effects demonstrates the model's robustness and its ability to satisfy various angular conditions. Based on the quantitative results, the HSL-CT model exhibits exceptional performance in characterizing leaf reflectance across varying incidence angles. Across the ten representative spectral bands displayed, the R^2 consistently exceed 0.95, demonstrating remarkably high fitting accuracy. Specifically, the 614.2 nm band achieves the optimal performance with an R^2 of 0.9979.

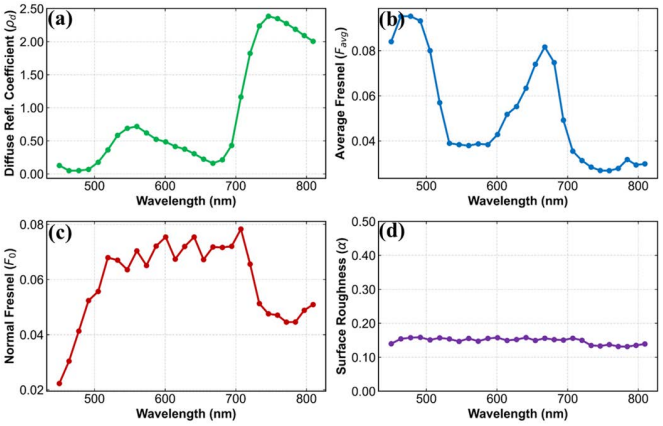


Fig. 8. Variation of HSL-CT model parameters across different wavelengths for potted plant leaves. (a) Diffuse reflectance ρ_d. (b) Average Fresnel reflectance F_{avg}. (c) Normal Fresnel reflectance F_0. (d) Surface roughnessα .

Even the 450.2 nm band, which had the lowest metric, still achieves an impressive R^2 of 0.9506. These outcomes validate the effectiveness of the HSL-CT model in simulating HSL reflectance for plant. Furthermore, the estimated parameters demonstrate strong generalization ability across continuous bands. This provides a reliable foundation for subsequent analysis of plant return characteristics.

Fig. 8 presents the estimated HSL-CT model parameters across different bands for the potted plant leaves. Fig. 8(a)–(d) represent the diffuse reflection coefficient (ρ_d), average Fresnel reflectance (F_{avg}), normal Fresnel reflectance (F_0), and surface roughness (α), respectively. The distribution of ρ_d aligns perfectly with plant reflectance theory. It exhibits a distinct green reflection peak (~550 nm) and a red absorption valley (~670 nm) in the visible spectrum. In the NIR region, ρ_d increases significantly, peaking around 750 nm. This pattern matches the typical spectral signature of healthy plants, characterized by low visible and high NIR reflectance. Analysis of the specular reflection parameters reveals that F_{avg} follows a trend nearly opposite to that of ρ_d. Specifically, F_{avg} shows a prominent valley at 550 nm and a peak at 670 nm, followed by a sharp decline in the NIR band. The normal Fresnel reflectance F_0 exhibits a multistage trend. It rises significantly between 450 and 500 nm, undergoes minor fluctuations from 500 to 700 nm, and declines before stabilizing after 700 nm. The minor fluctuations in the intermediate bands likely stem from environmental or instrumental noise during measurement. These behaviors highlight the wavelength dependence of Fresnel reflection and reflect complex differences in specular effects across the spectrum. Finally, the surface roughness parameter α remains extremely stable across all wavelengths, maintaining a value of approximately 0.15. Theoretically, surface roughness is an intrinsic physical property and should be wavelength-independent. However, the minor fluctuations observed during multiband model fitting are reasonable, likely arising from slight variations in raw data distribution across different bands. This stability aligns with the smooth surface characteristics of the potted leaves and further validates the rationality of our parameter estimation.

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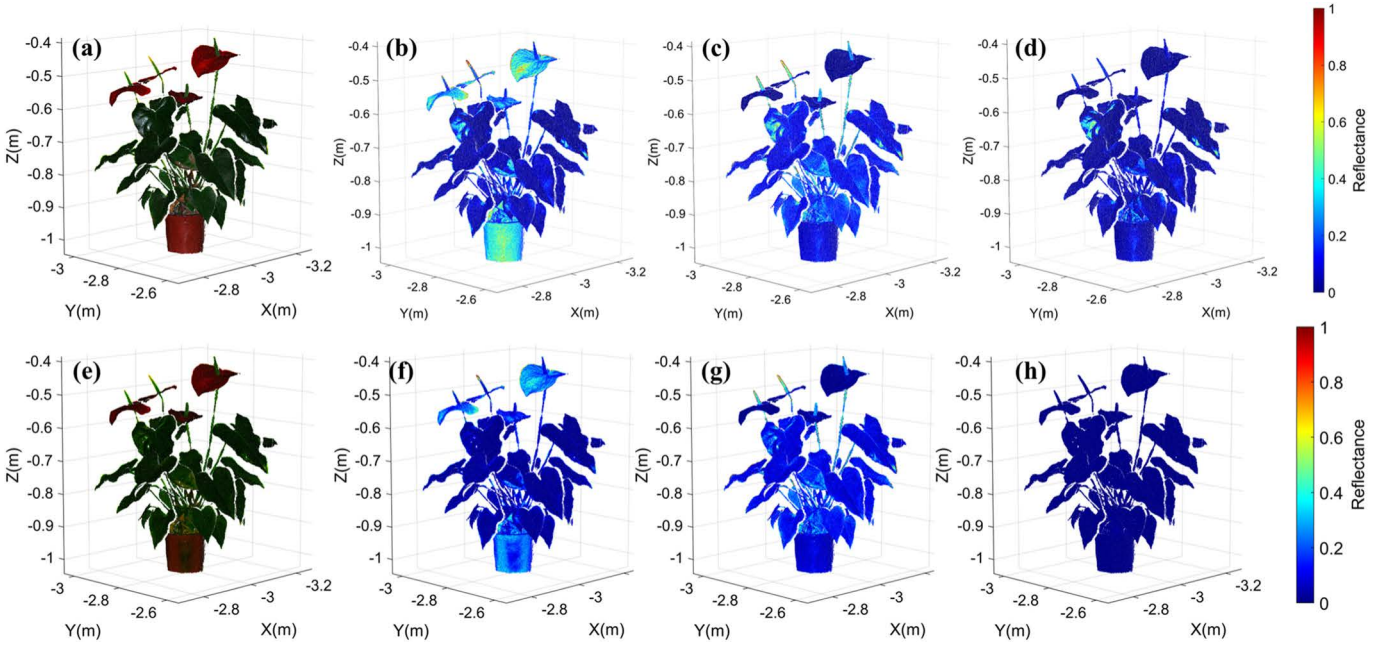


Fig. 9. Comparison of reflectance before and after correction based on the plant's global 3-D structural information. (a) Overall 3-D true-color (RGB) point cloud of the plant. (b)–(d) Uncorrected reflectance maps at 654.3, 546.3, and 450.2 nm, respectively. (e)–(h) Corresponding corrected reflectance at the respective wavelengths based on the global 3-D structural information.

Based on the HSL-CT model and the plant's global 3-D structural information, we performed correction on the reflectance of the entire plant. Fig. 9 compares the results before and after correction. The first row Fig. 9(a)–(d) presents the uncorrected data, while the second row Fig. 9(e)–(h) displays the corrected results. From left to right, the columns show the 3-D true-color (RGB) point cloud followed by reflectance visualizations at 654.3, 546.3, and 450.2 nm. All reflectance values are uniformly mapped to a color scale of 0 and 1 for intuitive comparison. Visualization results reveal that before correction, leaf regions facing the laser origin (areas with small incidence angles) exhibited abnormally high reflectance. These appear as yellow and light-blue highlights, particularly in subplot Fig. 9(b). After applying the correction based on the global 3-D structural information, these geometric effects were significantly mitigated. Postcorrection reflectance distributions across the canopy became highly uniform across all bands [Fig. 9(f)–(h)]. The results demonstrate that our method effectively removes false shadows and overly bright areas caused by distance and incidence angle effects. It restores the true spectral characteristics of the leaves, providing high-quality data for retrieving canopy biochemical parameters. Notably, the HSL-CT model parameters were primarily derived from green leaf data. Consequently, the red leaves at the top and the flower pot at the bottom may appear somewhat greenish in the globally corrected results. This is normal model behavior. We have fully accounted for this in our accuracy evaluation by excluding nongreen-leaf components from the final result.

C. Comparative Results Between the HSL-CT and LB Models

Table I presents the quantitative performance comparison between the proposed HSL-CT model and the LB model [23] across 28 spectral channels ranging from 450.2 to 809.3 nm using R^2 and *RMSE*. Across the entire spectrum, the HSL-CT model achieved higher fitting accuracy, with an average R^2 of 0.9935 compared to 0.9904 for the benchmark model, alongside a lower average *RMSE* of 0.0084.

To evaluate the performance of the proposed radiometric correction framework, spectral validation was conducted at four representative leaf points, namely Points 1–4, with varying incidence angles of 17°, 8°, 15°, and 10°, respectively (see Fig. 10). The reference reflectance spectra were acquired using a 300–1000 nm spectrometer equipped with an integrating sphere. Although the integrating sphere yields hemispherical reflectance while the corrected HSL represents specular-free directional diffuse reflectance, their absolute values are not expected to be numerically identical. The purpose of this comparison is to demonstrate that, after specular removal, the HSL reflectance spectra become significantly closer to the integrating-sphere reference in terms of spectral shape and relative band variations. As illustrated in the spectral comparison curves, the uncorrected raw HSL data exhibit significant positive radiometric distortion and spectral shape deviations compared to the spectrometer ground truth, yielding poor determination coefficients R^2 and high SAM values across all test points. Both correction models successfully mitigate these distortions and align closely with the ground truth in terms of overall spectral trajectory, demonstrating the effectiveness of accounting for geometric and structural effects. Specifically, compared with the uncorrected raw data, the LB model substantially improves accuracy, achieving R^2 values ranging from 0.6015 to 0.8723 alongside lower SAM values.

However, the proposed HSL-CT model consistently outperforms the LB model across all selected leaf samples. At Point 1 with an incidence angle of 17°, the HSL-CT model

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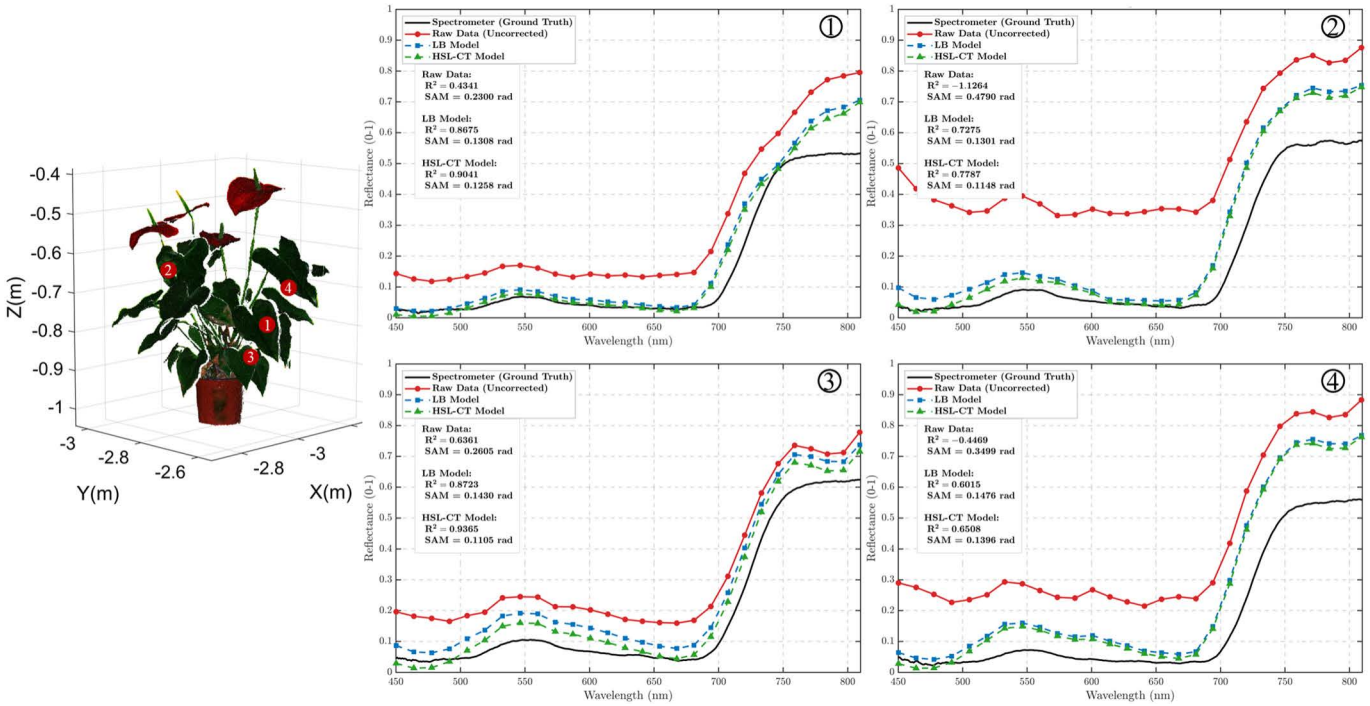


Fig. 10. Comparison of leaf reflectance spectra between uncorrected raw data, the LB model, and the proposed HSL-CT model against spectrometer ground truth at four representative leaf points (Points 1–4) with varying incidence angles (17°, 8°, 15°, and 10°, respectively).

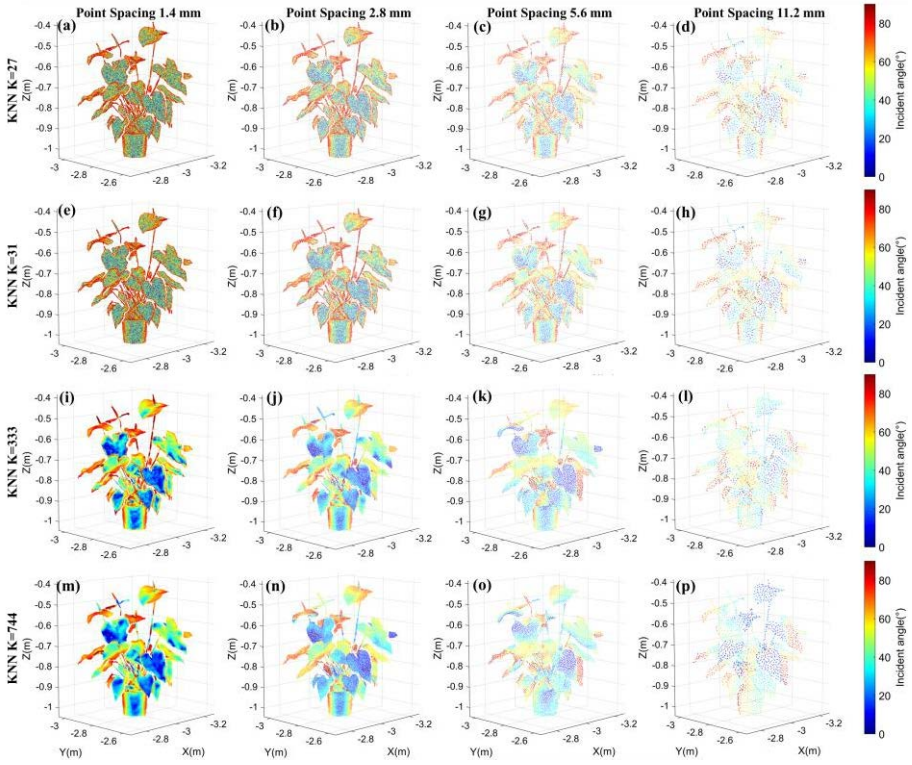


Fig. 11. Visualization results of incidence angles across 28 spectral bands under various point spacings and adaptive neighborhood point counts. Rows correspond to neighborhood sizes of K = 27, 31, 31, 333, and 744, respectively. (a), (e), (i), (m) Columns represent point spacings of 1.4 mm, (b), (f), (j), (n) 2.8 mm, (c), (g), (k), (o) 5.6 mm, and (d), (h), (l), (p) 11.2 mm. (a)–(d) Incidence-angle maps with K = 27. (e)–(h) Incidence-angle maps with K = 31. (i)–(l) Incidence-angle maps with K = 333. (m)–(p) Incidence-angle maps with K = 744. The color bar denotes incidence angle ranging from 0° to 80°. Severe edge artifacts appear for small K, while overly large K introduces spatial mixing across adjacent leaves under sparse point-cloud conditions.

increases R^2 from 0.8675 to 0.9041 and reduces the SAM from 0.1308 to 0.1258 rad. At Point 2 with an incidence angle of 8°, R^2 improves from 0.7275 to 0.7787, and the SAM decreases to 0.1148 rad. The most notable enhancement is

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TABLE I
QUANTITATIVE COMPARISON OF MODEL FITTING PERFORMANCE BETWEEN HSL-CT AND LB MODELS AGAINST MEASURED MULTIANGLE REFLECTANCE INTENSITY DATA ACROSS 14 LEAVES

Wavelength (nm)	R^2 (LB)	R^2 (HSL-CT)	*RMSE* (LB)	*RMSE* (HSL-CT)
450.2	**0.9553**	0.9506	**0.0070**	0.0073
463.9	0.9913	**0.9925**	0.0033	**0.0031**
477.7	0.9814	**0.9891**	0.0063	**0.0048**
491.5	0.9882	**0.9940**	0.0063	**0.0045**
505.2	0.9880	**0.9945**	0.0078	**0.0053**
518.9	0.9895	**0.9958**	0.0091	**0.0057**
532.6	0.9932	**0.9974**	0.0084	**0.0052**
546.3	0.9934	**0.9975**	0.0089	**0.0054**
559.9	0.9914	**0.9965**	0.0103	**0.0066**
573.5	0.9918	**0.9971**	0.0097	**0.0057**
587.1	0.9915	**0.9974**	0.0096	**0.0053**
600.7	0.9914	**0.9975**	0.0096	**0.0051**
614.2	0.9913	**0.9979**	0.0091	**0.0044**
627.6	0.9876	**0.9954**	0.0110	**0.0067**
641.0	0.9899	**0.9967**	0.0094	**0.0054**
654.3	0.9899	**0.9963**	0.0088	**0.0053**
667.6	0.9909	**0.9957**	0.0080	**0.0056**
680.9	0.9906	**0.9967**	0.0088	**0.0052**
694.0	0.9885	**0.9952**	0.0112	**0.0072**
707.2	0.9958	**0.9986**	0.0096	**0.0055**
720.2	**0.9980**	0.9975	**0.0081**	0.0090
733.2	**0.9957**	0.9943	**0.0132**	0.0152
746.1	**0.9935**	0.9917	**0.0166**	0.0188
758.9	**0.9938**	0.9919	**0.0159**	0.0183
771.6	**0.9938**	0.9920	**0.0156**	0.0177
784.3	**0.9949**	0.9927	**0.0137**	0.0165
796.8	**0.9952**	0.9930	**0.0131**	0.0158
809.3	**0.9948**	0.9927	**0.0133**	0.0157

observed at Point 3 with an incidence angle of 15°, where R^2 reaches a peak of 0.9365 compared to 0.8723 for the LB model, alongside a substantial SAM reduction to 0.1105 rad. Similarly, at Point 4 with an incidence angle of 10°, the HSL-CT model elevates R^2 to 0.6508 and lowers the SAM to 0.1396 rad. These quantitative improvements confirm that the HSL-CT model provides higher robustness and precision in correcting leaf reflectance under varying incidence angles.

IV. DISCUSSION

A. Impacts of Point Cloud Density on 3-D Canopy Geometry Estimation

Point spacings vary among different hardware acquisition systems. For instance, datasets collected in [26] and [27] feature a point density equivalent to approximately 5.5 mm point spacing, and point spacing directly determines the accuracy of global 3-D plant geometric information. Accordingly, we constructed a gradient of point spacings (1.4, 2.8, 5.6, 11.2, and 22.4 mm) to quantify their influences on incidence angle calculation and reflectance calibration. The neighborhood point count corresponding to each point spacing is determined using the point-density-adaptive neighborhood point selection strategy proposed in this study. Fig. 11 visualizes the incidence angles derived under different point spacings with their respective optimal neighborhood point counts. The 5.6 mm spacing represents the typical parameter adopted by common mainstream acquisition systems, while 1.4 mm corresponds to the native point spacing of the point cloud captured by our self-built measurement system. It can be observed that the optimal neighborhood point count of around 744 yields the most desirable incidence angle distribution under the 1.4 mm point spacing. Insufficient neighboring points lead to prominent leaf contour artifacts: the incidence angles at leaf edges become abnormally large, and the angular distribution over leaf interiors exhibits severe irregular fluctuations. As point spacing increases, fewer neighboring points are required to generate smooth incidence angle maps; nevertheless, fine local variations in leaf surface incidence angles cannot be adequately characterized. Taking the 5.6 mm point spacing as an example, obvious contour artifacts remain even at the optimal neighborhood count of 31. Although contour artifacts are eliminated when the neighborhood point count rises to 333, excessively sparse point distribution fails to capture subtle local variations of incidence angles across leaf surfaces. Therefore, with regard to the potted plants measured under the current experimental conditions, the point spacing needs to be controlled at 2.8 mm or below, among which the neighborhood point count set at 333 delivers the optimal outcomes for incidence angle acquisition.

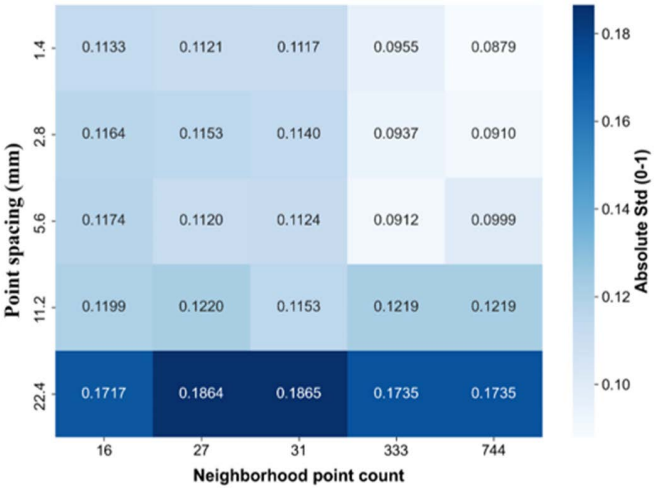


Fig. 12. Mean Std of calibrated reflectance over 28 spectral bands under different point spacings and adaptive neighborhood point counts.

Fig. 12 presents the quantitative statistical results of calibrated reflectance. Two distinct regularities can be observed in Fig. 12. First, under dense point cloud conditions with

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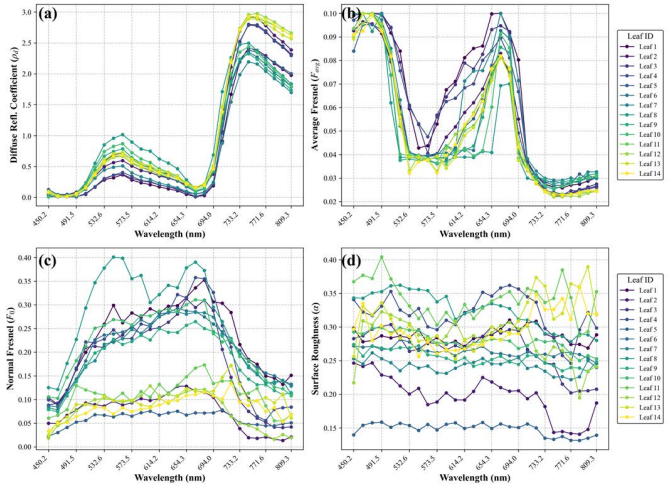


Fig. 13. Variation of HSL-CT model parameters across different wavelengths for 14 individual leaves. (a) Represents the diffuse reflection coefficient ρ_d. (b) Represents the average Fresnel reflectance F_{avg}. (c) Represents the normal Fresnel reflectance F_0. (d) Represents the surface roughness α. The curves with various colors indicate the parameter estimation results for leaf 1 through leaf 14.

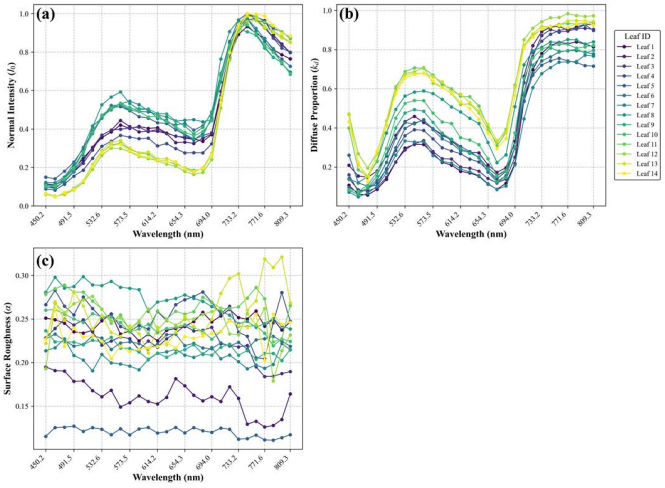


Fig. 14. Variation of LB model parameters across different wavelengths for 14 individual leaves. (a) Represents the normal backscattered intensity scaling factor I_0. (b) Represents the diffuse scattering proportion k_d. (c) Represents the surface roughness parameter α. The curves with various colors indicate the parameter estimation results for leaf 1 through leaf 14.

point spacings of 1.4 and 2.8 mm, the absolute Std gradually declines as the neighborhood point count increases from 16 to 744, hitting the minimum value of 0.0879 at the 1.4 mm point spacing with a neighborhood point count of 744. This indicates that high point density provides abundant local sampling information; expanding the neighborhood point count helps fit leaf surface normals more precisely, lowering the fluctuation of calibrated reflectance and boosting correction stability. Meanwhile, the optimization effect brought by enlarging the neighborhood gradually weakens when density slightly drops to 5.6 mm, where the overall absolute Std falls moderately, and the lowest value is only 0.0912. Second, the performance of neighborhood point selection is strongly constrained by point spacing, showing an obvious scale-dependent effect. Once point spacing reaches 11.2 and 22.4 mm (sparse point cloud range), continuously raising the neighborhood point count no longer reduces Std; on the contrary, absolute Std rises remarkably. Specifically, for the 11.2 mm point spacing group, the absolute Std climbs from 0.1199 to 0.1219 with the growth of neighborhood points. Under the sparsest 22.4 mm point spacing, all groups maintain high Std values above 0.17, peaking at 0.1865 when the neighborhood point count is 31. The underlying physical cause is that sparse point clouds demand an excessively large spatial search radius to gather enough neighboring points. Such an oversized radius will break the boundary of a single leaf, mixing points from neighboring leaves, branches, and background noise into normal vector calculation. Consequently, surface normal estimation is heavily distorted, the reflectance calibration result becomes unstable, and the Std increases sharply.

B. Analysis of Influences on Reflective Modeling

From the perspective of model construction logic, the LB model serves as a semi-empirical radiative model; see (4). It bundles microfacet distribution, geometric shadowing, and Fresnel refraction into a single exponential term with only two adjustable terms: diffuse fraction and surface roughness, without decoupling individual scattering mechanisms. By contrast, the proposed HSL-CT model is built on complete microfacet theory; see (5)–(10). It splits laser backscattering into four physically independent modules: diffuse base reflectance, microfacet normal distribution, geometric shadowing, and wavelength-dependent Fresnel reflection, and is simplified for the coaxial observation geometry unique to HSL. All HSL-CT parameters carry spectral attributes for separate quantification of each optical effect, while the LB model mixes all angle-related radiative losses in one empirical function. Results of 14 leaf samples plotted in Figs. 13 and 14 validate this structural gap. HSL-CT parameters evolve regularly across wavelengths and accurately recover typical plant spectral signatures such as green peaks, red valleys, and high near-infrared reflectance. The LB model's two parameters only show monotonous spectral trends and fail to capture scattering differences from varying leaf cuticles and microtopography, resulting in weak generalization over multiband multispecies leaf datasets.

Plant leaf surfaces feature micrometer micro-undulations whose specular intensity is jointly controlled by microfacet spatial distribution, inter-microstructure occlusion, and wavelength-variant refraction. The LB model cannot replicate the segmented angular scattering trend of sharp small-angle highlights and mild large-angle decay due to its oversimplified fitting strategy. The modular HSL-CT framework aligns fully with leaf optical physics: its microfacet function reproduces near-normal specular highlights, the shadow term estimates effective reflectance loss at steep angles, and band-sensitive Fresnel terms capture wavelength-specific spectral shifts. Fig. 15 compares R2 values across all bands. The LB model has wider R2 fluctuations and low-fitting troughs in visible bands, whereas HSL-CT yields higher average R2 with tighter bounds for more stable multiangle multiband fitting.

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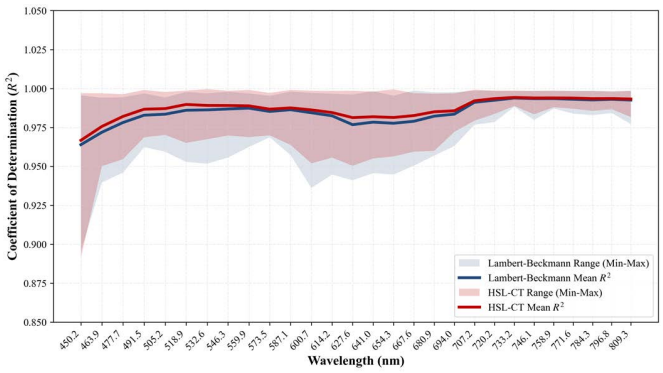


Fig. 15. Comparison of the R^2 between the HSL-CT model and the LB model across different wavelengths. The solid lines represent the mean R^2 values across the 14 individual leaves, while the corresponding shaded areas indicate the variation range bounded by the minimum and maximum R^2 values.

Combined analysis of Figs. 13 and 15 shows HSL-CT has four physical tunable parameters to match epidermal and wax variations across the 14 leaves, whereas the LB model lacks enough parametric freedom to resolve species and wavelength scattering disparities. In correction processing, see (12) and (13); HSL-CT calculates band-wise specular proportions to fully remove geometry-induced spectral bias and retrieve true leaf reflectance. Limited by simplified physical premises, the LB model leaves obvious spectral distortion in highlight zones and delivers weaker improvement in SAM and RMSE metrics.

C. Future Work

Although the HSL-CT model and whole-plant 3-D structural correction framework deliver satisfactory performance for green leaf tissues, this study still bears noticeable limitations worthy of further exploration. It is worth emphasizing that the present work serves as a preliminary feasibility validation of the proposed correction framework under controlled experimental conditions, and the efficacy and generalizability of the method have not been fully verified in complex and diverse field scenarios. The HSL-CT parameter library is calibrated solely based on green leaf samples, which introduces residual spectral deviations when correcting nongreen canopy components such as flowers and branches. The current unified modeling scheme does not distinguish optical scattering differences among various plant tissues, limiting its generalization ability for heterogeneous complex canopies.

Future research will focus on mitigating the above limitations. A spectral preclassification module will be integrated into the correction pipeline to differentiate distinct plant tissues and background materials before reflectance modeling. Targeted tissue-adaptive scattering model parameters will then be generated to reduce cross-tissue spectral offsets and further boost correction accuracy. Moreover, we will expand the diversity of plant samples and experimental conditions to systematically validate the transferability and robustness of the proposed framework across variable field environments, thereby promoting its practical application in field-scale quantitative plant remote sensing.

V. Conclusion

This study proposes a canopy-scale reflectance correction framework for HSL utilizing global 3-D plant structural information to mitigate incidence angle artifacts. First, an adaptive neighborhood selection algorithm is developed to calculate surface normals and incidence angles. Optimized iteratively via point cloud density and the CGC metric, the method eliminates over-smoothing and noise amplification from fixed neighborhood sizes, and alleviates geometric distortion in normal estimation on sparse discrete canopy point clouds. Second, a physically refined HSL-CT model is constructed to separate diffuse and specular reflectance across 28 spectral bands (445–810 nm). The HSL-CT model attains an average fitting R^2 above 0.96 for all bands, with a peak value of 0.993 in the near-infrared range, and outperforms the LB benchmark model with lower RMSE. Taking integrating-sphere spectrometer measurements at four leaf positions with varying incidence angles as ground truth, we verify the correction performance. The HSL-CT model yields higher R^2 and smaller SAM values than the LB model for all test points. After correction, the visible-band reflectance RMSE drops from over 0.4 to roughly 0.1; angular specular over-brightness is eliminated, restoring intrinsic plant spectra. The proposed framework overcomes drawbacks of conventional single-point correction methods and provides incidence-angle-insensitive reflectance values for accurate plant biochemical parameter inversion.

Nevertheless, the method has obvious limitations. The HSL-CT parameter library is only calibrated with green leaves, leading to residual spectral biases when correcting nongreen tissues like branches and flowers. In future research, we will add a tissue spectral preclassification module to establish tissue-specific scattering parameters, and expand datasets of various plant types and field scenarios to validate the generalization ability of our framework, so as to promote the application of HSL in precision agriculture and ecological monitoring.

References

[1] S. Chen, C. Song, X. Mao, and S. Shang, "Modeling response of spring wheat yield to soil water and salt contents and its application in scheduling brackish water irrigation," *Comput. Electron. Agricult.*, vol. 200, Sep. 2022, Art. no. 107216, doi: 10.1016/j.compag.2022.107216.

[2] H. Ma and S. Liang, "Development of the GLASS 250-m leaf area index product (version 6) from MODIS data using the bidirectional LSTM deep learning model," *Remote Sens. Environ.*, vol. 273, May 2022, Art. no. 112985, doi: 10.1016/j.rse.2022.112985.

[3] D. Song et al., "Development of crop chlorophyll detector based on a type of interference filter optical sensor," *Comput. Electron. Agricult.*, vol. 187, Aug. 2021, Art. no. 106260, doi: 10.1016/j.compag.2021.106260.

[4] P. Putkiranta et al., "The value of hyperspectral UAV imagery in characterizing Tundra vegetation," *Remote Sens. Environ.*, vol. 308, Jul. 2024, Art. no. 114175, doi: 10.1016/j.rse.2024.114175.

[5] Y. Zhang, M. Migliavacca, J. Penuelas, and W. Ju, "Advances in hyperspectral remote sensing of vegetation traits and functions," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112121, doi: 10.1016/j.rse.2020.112121.

[6] E. Gril et al., "Using airborne LiDAR to map forest microclimate temperature buffering or amplification," *Remote Sens. Environ.*, vol. 298, Dec. 2023, Art. no. 113820, doi: 10.1016/j.rse.2023.113820.

[7] W. Ni-Meister, A. Rojas, and S. Lee, "Direct use of large-footprint LiDAR waveforms to estimate aboveground biomass," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113147, doi: 10.1016/j.rse.2022.113147.

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[8] B. Fu et al., "Quantifying vegetation species functional traits along hydrologic gradients in Karst wetland based on 3D mapping with UAV hyperspectral point cloud," *Remote Sens. Environ.*, vol. 307, Jun. 2024, Art. no. 114160, doi: 10.1016/j.rse.2024.114160.
[9] D. Han et al., "Estimation of canopy fAPAR using optical reflectance and airborne LiDAR data," *Remote Sens. Environ.*, vol. 332, Jan. 2026, Art. no. 115065, doi: 10.1016/j.rse.2025.115065.
[10] T. Hakala, J. Suomalainen, S. Kaasalainen, and Y. Chen, "Full waveform hyperspectral LiDAR for terrestrial laser scanning," *Opt. Exp.*, vol. 20, no. 7, p. 7119, Mar. 2012, doi: 10.1364/oe.20.007119.
[11] G. Wei et al., "Development and prospect of hyperspectral LiDAR for Earth observation," *Nat. Remote Sens. Bull.*, vol. 25, no. 1, pp. 501–513, Mar. 2021, doi: 10.11834/jrs.20210086.
[12] P. Ray, D. Salido-Monzú, S. L. Camenzind, and A. Wieser, "Supercontinuum-based hyperspectral LiDAR for precision laser scanning," *Opt. Exp.*, vol. 31, no. 20, pp. 33486–33499, Sep. 2023, doi: 10.1364/oe.498576.
[13] A. Wang et al., "Potential of hyperspectral LiDAR in individual tree segmentation: A comparative study with multispectral LiDAR," *Urban Forestry Urban Greening*, vol. 104, Feb. 2025, Art. no. 128658, doi: 10.1016/j.ufug.2024.128658.
[14] A. Wang et al., "Integration of LiDAR and hyperspectral imagery for tree species identification at the individual tree level," *Photogrammetric Rec.*, vol. 40, no. 190, p. 70007, Apr. 2025, doi: 10.1111/phor.70007.
[15] L. Xu et al., "Mapping 3D plant chlorophyll distribution from hyperspectral LiDAR by a leaf-canopyradiative transfer model," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103649, doi: 10.1016/j.jag.2024.103649.
[16] J. Bai et al., "An exploration, analysis, and correction of the distance effect on terrestrial hyperspectral LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 60–83, Apr. 2023, doi: 10.1016/j.isprsjprs.2023.03.001.
[17] E. N. Ramtvedt, O. M. Bollandsås, E. Næsset, and T. Gobakken, "Relationships between single-tree mountain birch summertime albedo and vegetation properties," *Agricult. Forest Meteorol.*, vol. 307, Sep. 2021, Art. no. 108470, doi: 10.1016/j.agrformet.2021.108470.
[18] P. Tao et al., "Recognition of ecological vegetation fairy circles in intertidal salt marshes from UAV LiDAR point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103029, doi: 10.1016/j.jag.2022.103029.
[19] D. Krafft, C. G. Scarboro, W. Hsieh, C. Doherty, P. Balint-Kurti, and M. Kudenov, "Mitigating illumination-, leaf-, and view-angle dependencies in hyperspectral imaging using polarimetry," *Plant Phenomics*, vol. 6, p. 0157, Mar. 2024, doi: 10.34133/plantphenomics.0157.
[20] Y. Li and J. Huang, "Remote sensing of pigment content at a leaf scale: Comparison among some specular removal and specular resistance methods," *Remote Sens.*, vol. 11, no. 8, p. 983, Apr. 2019, doi: 10.3390/rs11080983.
[21] B. Yang, Y. Knyazikhin, H. Zhao, and Y. Ma, "Contribution of leaf specular reflection to canopy reflectance under black soil case using stochastic radiative transfer model," *Agricult. Forest Meteorol.*, vol. 263, pp. 477–482, Dec. 2018, doi: 10.1016/j.agrformet.2018.08.024.
[22] R. Gaulton, F. M. Danson, F. A. Ramirez, and O. Gunawan, "The potential of dual-wavelength laser scanning for estimating vegetation moisture content," *Remote Sens. Environ.*, vol. 132, pp. 32–39, May 2013, doi: 10.1016/j.rse.2013.01.001.
[23] J. Bai et al., "A novel algorithm for leaf incidence angle effect correction of hyperspectral LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 9404348, doi: 10.1109/TGRS.2021.3070652.
[24] X. Qian et al., "Analyzing the effect of incident angle on echo intensity acquired by hyperspectral LiDAR based on the lambert-beckman model," *Opt. Exp.*, vol. 29, no. 7, p. 11055, Mar. 2021, doi: 10.1364/oe.420468.
[25] H. Zhang et al., "Analyzing and correcting the incidence angle effect of the hyperspectral LiDAR echo intensity," *Opt. Laser Technol.*, vol. 193, Jan. 2026, Art. no. 114224, doi: 10.1016/j.optlastec.2025.114224.
[26] P. Hu et al., "Tree trunk disease mapping using hyperspectral LiDAR," *Comput. Electron. Agricult.*, vol. 239, Dec. 2025, Art. no. 111005, doi: 10.1016/j.compag.2025.111005.
[27] J. Bai et al., "Edge effect issue on hyperspectral LiDAR point cloud data: Generation principle and framework of detection and filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5702610, doi: 10.1109/TGRS.2026.3699269.
[28] B. Qi, G. Yang, Y. Zhang, and C. Wang, "Target intensity correction method based on incidence angle and distance for a pulsed LiDAR system," *Appl. Opt.*, vol. 63, no. 10, p. A86, Apr. 2024, doi: 10.1364/ao.505690.
[29] S. Shi, Q. Xu, C. Gong, W. Gong, X. Tang, and B. Zhou, "Analysis, simulation, and scanning geometry calibration of Palmer scanning units for airborne hyperspectral light detection and ranging," *Remote Sens.*, vol. 17, no. 8, p. 1450, Apr. 2025, doi: 10.3390/rs17081450.
[30] S. Shi et al., "Waveform information accurate extraction for massive and complex waveform data of hyperspectral LiDAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1020–1038, 2025, doi: 10.1109/JSTARS.2024.3495039.
[31] H. Shao et al., "Feasibility study of wood-leaf separation based on hyperspectral LiDAR technology in indoor circumstances," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 729–738, 2022, doi: 10.1109/JSTARS.2021.3134651.
[32] M. Sabzali and L. Pilgrim, "A comprehensive review of mathematical error characterization and mitigation strategies in terrestrial laser scanning," *Remote Sens.*, vol. 17, no. 14, p. 2528, Jul. 2025, doi: 10.3390/rs17142528.
[33] L. Deng et al., "Leaf bidirectional reflectance distribution function (BRDF) prediction with phenotypic traits in four species: Development of a novel measuring and analyzing framework," *Plant Phenomics*, vol. 7, no. 4, Dec. 2025, Art. no. 100135, doi: 10.1016/j.plaphe.2025.100135.



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